SUPPORTING ADULT GUIDANCE

Area of parallelograms and trapezia

Perimeter and Area · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Shear Slider, learners drag the corners of a parallelogram on a 1 m grid; sliding its top edge sideways (a shear) changes the slant but not the base, height or area. Slide the triangle cuts a right-angled triangle from one end and slides it to the other, making a rectangle. In trapezium mode, learners set the two parallel sides and the height; Add a turned copy brings in a copy turned half a turn, making a parallelogram with base a + b. Every corner is on a grid point, so areas are exact (halves shown as ½). Six questions and a stretch on the trapezium formula follow, then quick areas, true-or-not statements and shapes made to order.

Driving Question: *Can a shape be rearranged into one we already know?*

The mathematics, in plain terms

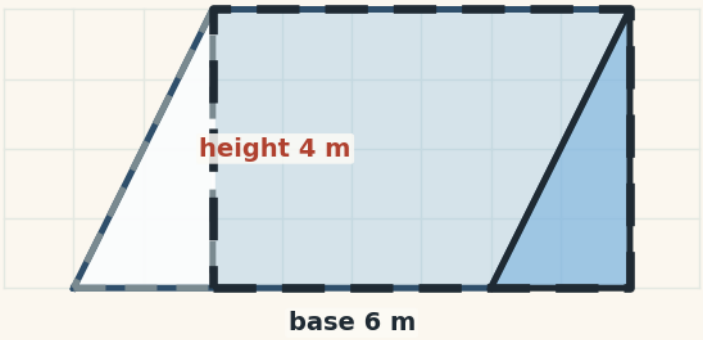
Area of a parallelogram = base × perpendicular height (it is a sheared rectangle; shearing preserves area). Area of a trapezium = ½ × (a + b) × h, where a and b are the parallel sides: two congruent trapezia, one rotated 180° about the midpoint of a leg, form a parallelogram of base a + b. These are the same half-turn and slide moves used in Lesson 2 and in Angles Lesson 3.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A parallelogram becomes a rectangle

Find the area of a parallelogram with base 6 m and height 4 m.

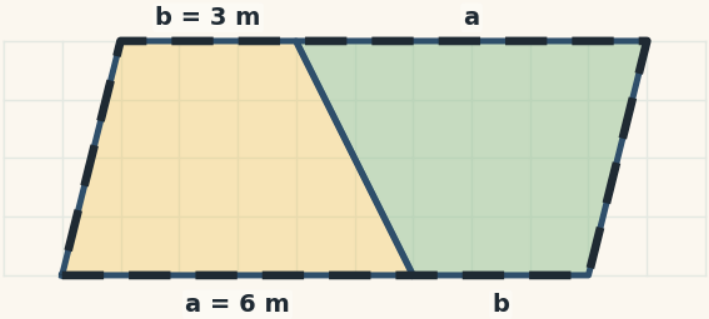


1. Cut off the triangle at one end.	a right-angled triangle
2. Slide it to the other end.	a 6 m by 4 m rectangle
3. Area of the rectangle	$6 \times 4 = 24$ m^2

Area = base × height = 24 m^2.

Example 2 — Two trapezia make a parallelogram

Find the area of a trapezium with parallel sides 6 m and 3 m, and height 4 m.



1. Add a copy turned half a turn.	a parallelogram
2. Its base is a + b.	6 + 3 = 9 m
3. The parallelogram: 9 × 4	36 m^2
4. The trapezium is half of it.	18 m^2

Area = ½ × (6 + 3) × 4 = 18 m^2.

Rearrange first, then the formula

Learners slide and turn pieces before any formula appears, so each formula describes a picture they have made. The stretch makes the trapezium argument explicit.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Shear Slider: parallelograms, then trapezia, with six questions. Stretch: why ½ × (a + b) × height.	~16 min
Movement	Quick areas.	~7 min
Reflection	True or not? Formulas from rearranging.	~7 min
Creativity	Make a shape with a given area.	~10 min
Rest	Lean it over; the space stays the same.	~2 min
Nutrition	Intellectual: seeing differently.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Multiplying the base by the slanted side.
Ask: “After sliding the triangle, which length is the side of the rectangle?”

Multiplying the two parallel sides of a trapezium.
Ask: “What does a turned copy make? What is that shape's base?”

Forgetting to halve for a trapezium.
Ask: “How many trapezia make the parallelogram?”

Thinking a leaning parallelogram holds less than a rectangle with the same base and height.
Ask: “Where does the triangle you cut off go?”

Using a non-perpendicular height.
Ask: “Is that line at right angles to the base?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: area of rectangles and triangles (Lesson 2). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says a trapezium with parallel sides 3 m and 5 m and height 2 m has an area of 15 m^2. The aim is a question that brings back the picture.

- “Where did the 15 come from?”

- “What would two copies of the trapezium make?”
- “What is the base of that shape?”
- “So what is half of it?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Do learners need to memorise the trapezium formula?
It helps, but the picture (two copies make a parallelogram) lets them rebuild it at any time.

Why only whole-metre corners?
So every area is exact; halves appear naturally for trapezia and are shown as ½.

Is the stretch required?
No. The secure core is base × height for parallelograms and ½ × (a + b) × height for trapezia, with the perpendicular height.

What comes next?
Lesson 4: composite shapes and design choices.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.